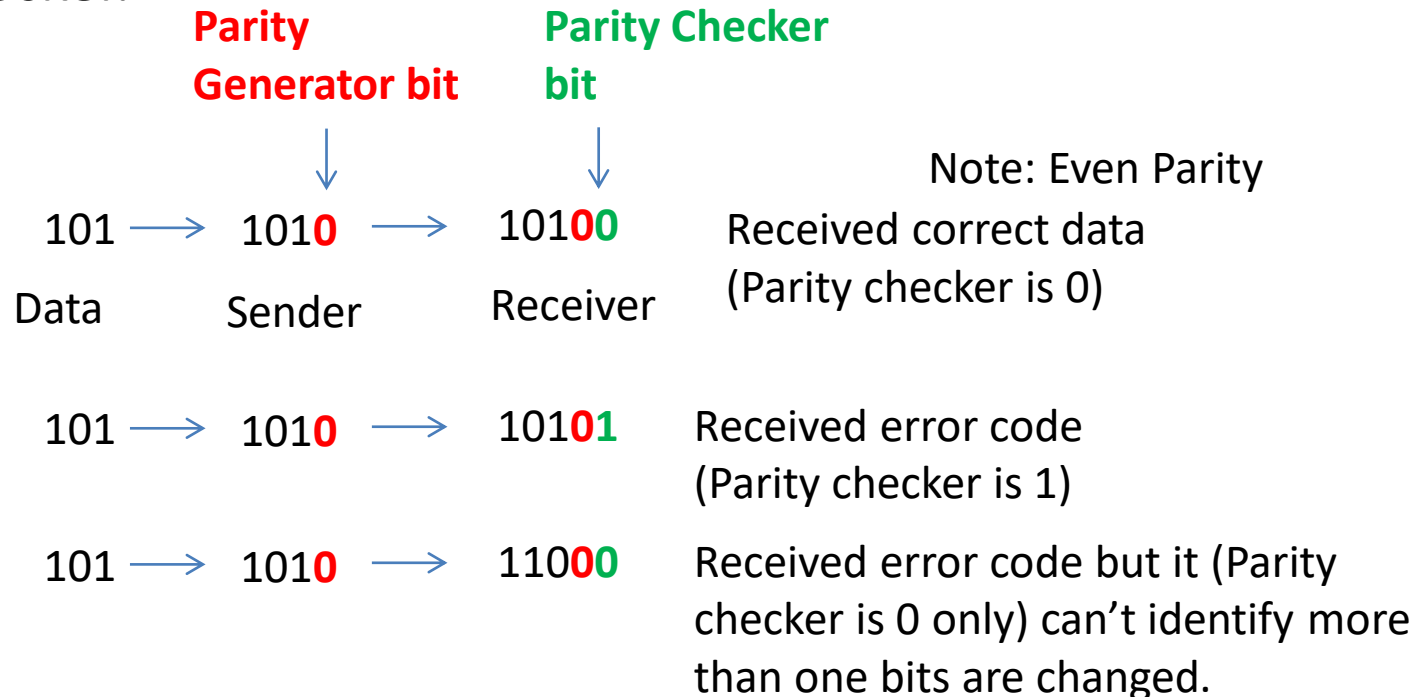


Parity Generation and Checking

XOR function useful for error detection and correction.

Transmit n-bit data with an additional parity bit.

Any faulty bit in the data will be detected by a parity checker.



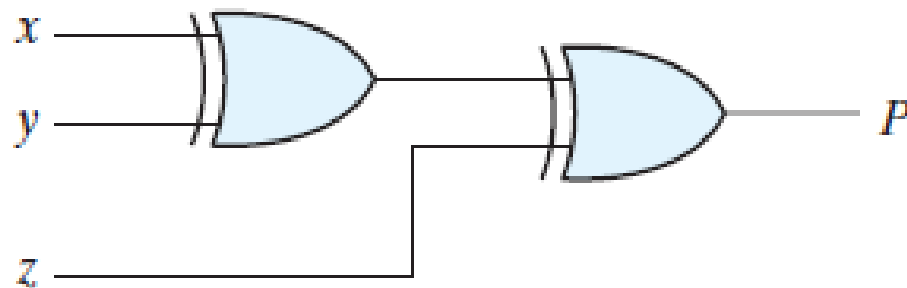
Even-Parity-Generator Truth Table

Three-Bit Message			Parity Bit
x	y	z	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Note: **Even parity generator** means it generates "1" when input data has odd number of "1" or else it generates "0"

Note: **Odd parity generator** means it generates "1" when input data has even number of "1" or else it generates "0".

This is complement of Even parity generator.



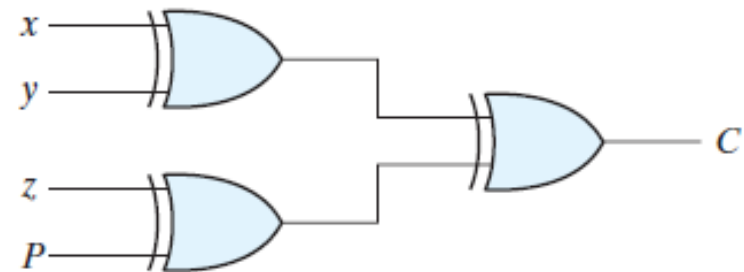
(a) 3-bit even parity generator

Parity Checker

Even-Parity-Checker Truth Table

Four Bits Received				Parity Error Check
x	y	z	P	C
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

$$C = x \oplus y \oplus z \oplus P$$



(b) 4-bit even parity checker

Note: **Even parity checker** means it generates “1” when input data(Including parity generator bit) has odd number of “1” or else it generates “0”.

If **output is “1”, the data has error** (or) it is corrupted data.

Note: **Odd parity checker is complement Of Even parity checker.**

K – map for Even Parity Generator & Checker

yz \ x	00	01	11	10
0	0	1	0	1
1	1	0	1	0

P (Even parity generator)

AB \ CP	00	01	11	10
00	0 ⁰	1 ¹	0 ³	1 ²
01	1 ⁴	0 ⁵	1 ⁷	0 ⁶
11	0 ¹²	1 ¹³	0 ¹⁵	1 ¹⁴
10	1 ⁸	0 ⁹	1 ¹¹	0 ¹⁰

P (Even parity Checker)

$$= \bar{A} \bar{B} (\bar{C} D + \underline{C} \bar{D}) + \bar{A} B (\bar{C} \bar{D} + C D) + A B (\bar{C} D + C \bar{D}) + A \bar{B} (\bar{C} \bar{D} + C D)$$

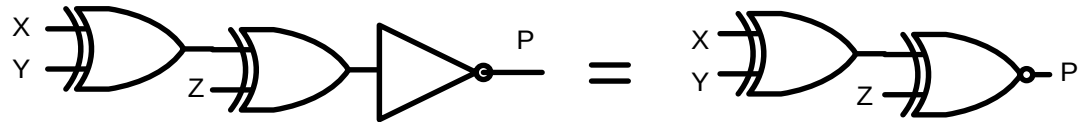
$$= \bar{A} \bar{B} (C \oplus D) + \bar{A} B (\overline{C \oplus D}) + A B (C \oplus D) + A \bar{B} (\overline{C \oplus D})$$

$$= (\bar{A} \bar{B} + A B) (C \oplus D) + (\bar{A} B + \underline{A} \bar{B}) (\overline{C \oplus D})$$

$$= (A \oplus B) \oplus (C \oplus D)$$

Odd Parity Generator

X	Y	Z	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0



$$P = \overline{X \oplus Y \oplus Z}$$

Odd Parity Checker

Draw the truth table and Verify yourself. Draw the circuit

$$C = \overline{X \oplus Y \oplus Z \oplus P}$$